LE NGUYEN DUY LUAN

Mobile Developer | On-device AI / Computer Vision

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Portfolio | GitHub | LinkedIn

Professional Summary

Motivated and detail-oriented Mobile Developer specializing in Native Android (Kotlin, Jetpack Compose) and On-device AI/Computer Vision (TensorFlow Lite). Passionate about building performance-optimized, offline-first mobile applications with real-time ML capabilities. Possesses strong problem-solving skills, solid understanding of fullstack architectures, and a dedication to writing clean, maintainable code.

Education

Nha Trang University Graduated: Mar 2026
Bachelor of Information Technology *Nha Trang, Vietnam*

- **GPA**: 7.94/10 (3.25/4.0)
- **Graduation Project**: Lung X-ray Disease Classification using Segmentation-guided Masking — Score: 9.1/10
- **Relevant Coursework**: Data Structures & Algorithms, Mobile Device Programming, Software Architecture & Design, Image Processing, Artificial Intelligence

Technical Skills

Languages: Java, Kotlin, JavaScript, Python, SQL
Mobile: Android SDK, Jetpack Compose, CameraX, Room DB, WorkManager, Coroutines & Flow
AI / Computer Vision: TensorFlow Lite, MediaPipe, MLKit (Pose Estimation), YOLO, OpenCV, Gemini API
Web / Backend: Firebase (Auth, Firestore, Storage), Supabase, REST API design, OkHttp
Tools: Git/GitHub, Android Studio, Gradle, ADB (Android Debug Bridge), KSP, VS Code, Figma (basic)
AI-Assisted Dev: Google Antigravity, OpenAI Codex
Languages (Spoken): Vietnamese (native), English (technical reading/writing)

Projects

VocabLensAI — Native Android English Learning Platform | *Personal Project*

- **Tech Stack**: Kotlin, Jetpack Compose, Room DB, CameraX, TensorFlow Lite, Gemini API, Firebase, Supabase.
- Developed a native offline-first Android app (Kotlin, Compose, MVVM) integrating CameraX and a lightweight TensorFlow Lite model to instantly convert real-world surroundings into vocabulary flashcards.
- Addressed the vocabulary retention problem by designing an SM-2 Spaced Repetition engine that schedules daily reviews at optimal intervals, using a hybrid Room/Firestore background progress sync.
- Built a real-time PvP Arena using Firebase Firestore with an ELO rating calculator, alongside a Chess.com-style game review UI displaying match accuracy and dynamic AI feedback.

Lung X-ray Disease Classification | *Graduation Project, Nha Trang University*

- **Tech Stack**: Python, TensorFlow/Keras, OpenCV, U-Net (Segmentation), ResNet/DenseNet (Classification).
- Solved the classification noise problem by developing a U-Net segmentation pipeline to isolate lung regions in chest X-ray images, boosting diagnostic accuracy for ResNet/DenseNet classifiers (Score: 9.1/10).
- Documented the end-to-end medical imaging pipeline from pre-processing to Grad-CAM heatmap visualization, preparing the project for clinical research handoffs.

AI Fitness Coach (Android) | *Personal Project*

- **Tech Stack**: Kotlin, Jetpack Compose, MediaPipe/MLKit (Pose Estimation), Room DB, CameraX.
- Addressed the injury risk of unsupervised home workouts by implementing camera-based pose estimation (MediaPipe/MLKit) to extract joint coordinates and calculate angles in real-time.
- Built a custom rule-based analysis engine providing instant visual corrective feedback (e.g., ”Keep your back straight”) and automated rep counting, caching history 100% offline via Room DB.